

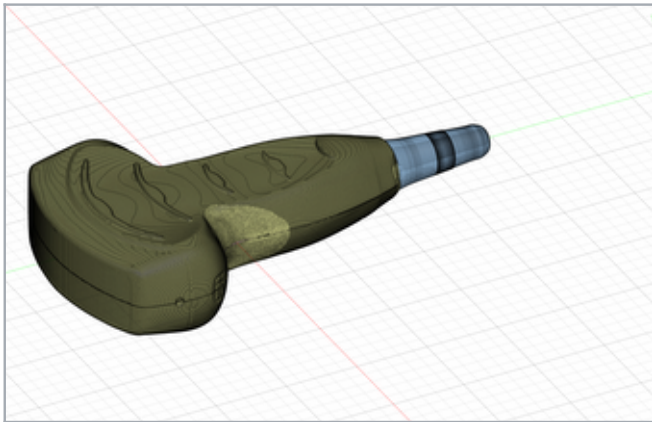
(a) Instrumented probe build

1 Dimensional survey



GE 4C-RS convex array · steel rule + digital protractor

2 CAD reconstruction



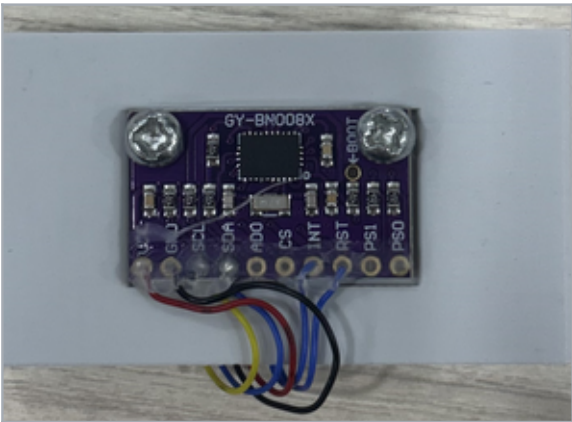
Fusion 360 mesh · ultrasound_probe v5

3 Printed clamp shell



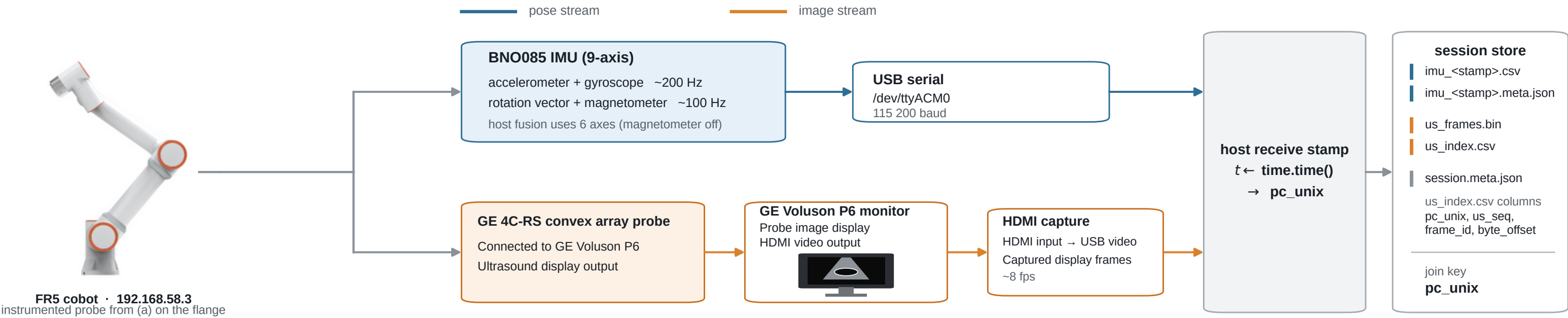
hinged split shell + sensor bracket

4 IMU on bracket

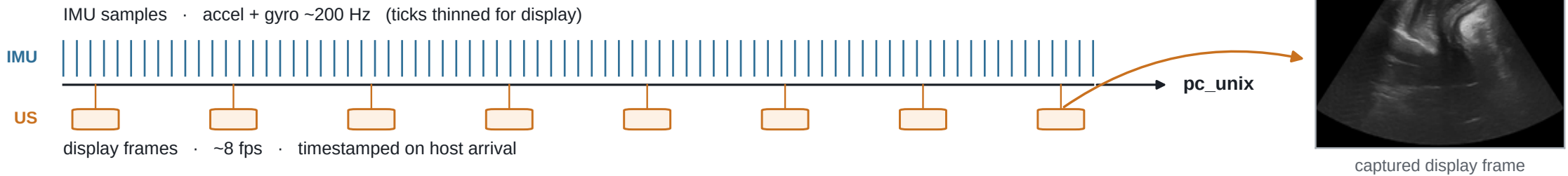


BNO085 breakout (GY-BNO08X), I²C to MCU

(b) Acquisition architecture — two paths, one clock



(c) Common time base



Arrival-time alignment

IMU: USB serial
Ultrasound: monitor display → HDMI capture
Host timestamps both streams at arrival
Display and capture latency measured separately